

Wildlife Image Processing & Semantic Search System

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Agenda

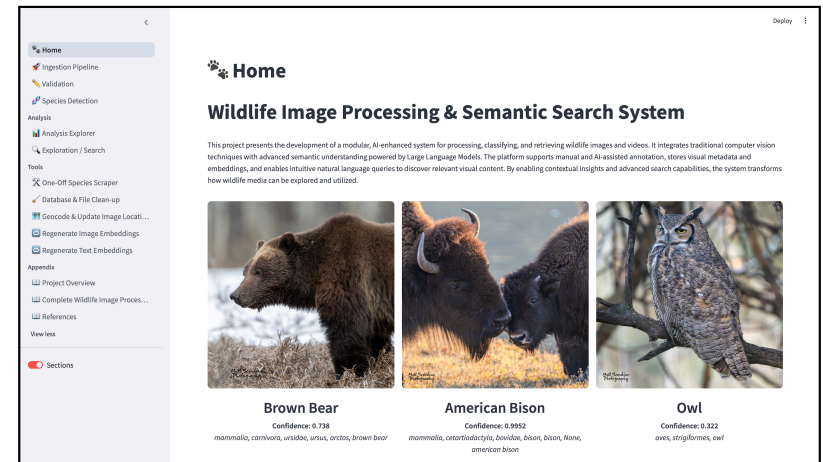
* Why This Project

* The Hummingbird

* Architecture

* Demo

* Q&A



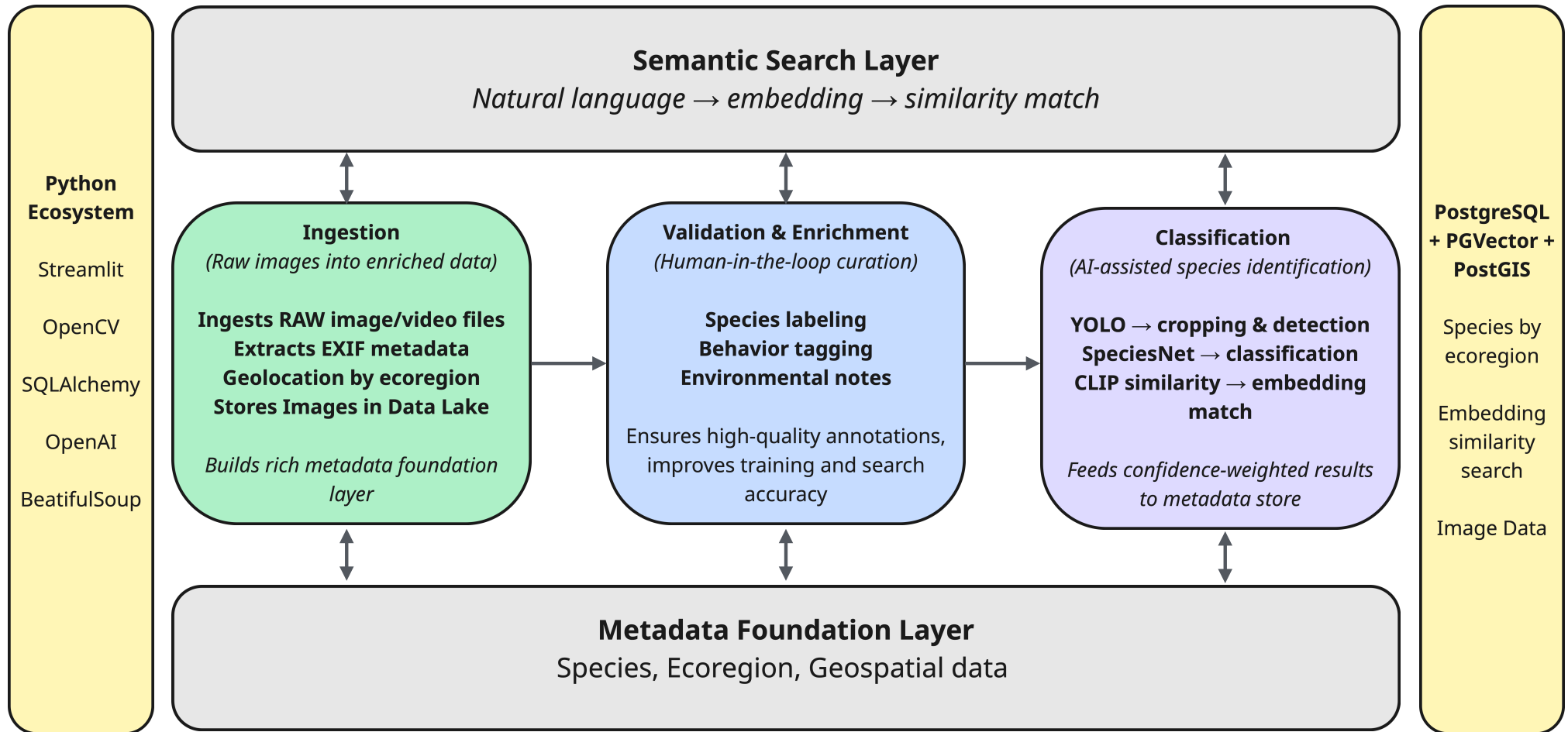
Link to Hummingbird: <https://vimeo.com/1095183654>



Wildlife Image Processing & Semantic Search System Architecture



From Field Capture to Searchable Insight



Link to Demo: <https://vimeo.com/1095187131>

Summary

Every image is a question. This system helps find the answer

- * Built a complete wildlife image processing system to turn photos into insights
- * Combined AI, spatial data, and semantic search to make observations smarter
- * Grounded species predictions in real-world ecology—location, behavior, habitat
- * Enables faster discovery of patterns in migration, seasonal changes, and biodiversity
- * Modular and extensible: adapts to new species, regions, and conservation needs
- * Designed for anyone documenting nature—photographers, researchers, citizen scientists

Q & A